

A Risk-Driven Cyber Threat Intelligence Framework using Graph-based Analysis and Zero-Trust Sharing

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Abstract—Cyber threats are becoming increasingly sophisticated, creating major challenges for traditional security monitoring systems. Organizations require advanced cyber-threat intelligence mechanisms capable of detecting complex attack patterns from large volumes of security telemetry. This study proposes a risk-driven cyber-threat intelligence framework that integrates telemetry collected from EDR/XDR sensors, a Unified Data Model (UDM), and Graph Neural Network (GNN)-based analysis for effective threat detection. Initially, heterogeneous security logs are normalized and mapped into the UDM to ensure consistent data representation. A behavioural graph is then constructed to represent relationships among entities such as users, processes, files, and network connections. The proposed GNN model performs relational learning on this graph to identify malicious behaviours and classify cyber threats. Experimental evaluation demonstrates that the framework improves detection accuracy while reducing false alarms compared with conventional machine learning methods. The proposed approach enhances the capability to detect complex and multi-stage cyber-attacks in modern enterprise security environments.

Keywords—Cyber Threat Intelligence, Malware Detection, Sigma Normalisation, Graph Neural Networks, Attack Graphs, and Threat Classification

I INTRODUCTION

The pace at which cyber threats are evolving has exceeded that of the past, and modern digital infrastructures are becoming increasingly interconnected, data-driven, and dependent on cloud technologies [1]. Malware, phishing, ransomware, insider attacks, and advanced persistent threats (APTs) continue to exploit network and endpoint vulnerabilities, posing significant risks to organizations [2]. Cyber Threat Intelligence (CTI) has therefore become essential for identifying attackers' intentions, predicting their future actions, and enabling the development of proactive defense strategies [3]. The effectiveness of CTI relies on continuous collection, analysis, and management of diverse security data to generate actionable insights that support timely decision-making [4].

The increasing complexity of cyber threats is driven by factors such as widespread cloud adoption, remote access systems, large-scale IoT deployments, and the use of AI-enabled automation and advanced obfuscation techniques by attackers [5]. Additionally, fragmented security systems, heterogeneous data types, and vast volumes of raw telemetry logs create security blind spots, while limited inter-organizational data sharing and financially motivated cybercrime groups further complicate threat detection [6]. Current CTI methods mainly rely on log correlation, signature-based detection, anomaly monitoring, and machine learning for malware classification [7]. Moreover, blockchain-based NLP-powered extraction lacks sufficient contextual depth for multi-stage attack representation.

However, existing CTI approaches face limitations in integrating multi-source data, accurately correlating complex attack patterns, and prioritizing threats in real time, motivating the need for a unified, context-aware intelligence workflow. To address these gaps, this study aims to develop a unified CTI workflow that integrates risk-based threat modelling, automated telemetry gathering, Sigma-driven normalization, and graph-based threat analysis. The approach improves correlation accuracy, reveals hidden attack linkages, and enables secure intelligence sharing through zero-trust-based mechanisms and TLP enforcement.

The primary contributions of this work are:

- ✓ An integrated CTI procedure combining MITRE-STRIDE risk assessment, EDR/XDR data collection, and Sigma normalization for unifying multi-source malware analysis.
- ✓ A graph-oriented detection framework employing GNNs and attack graph reasoning to accurately classify malware-related threats with minimal misclassification.

- ✓ A high-performance evaluation demonstrating robustness, scalability, and near-perfect detection metrics in real-world scenarios.

This unified CTI workflow, combining risk assessment, automated telemetry, Sigma normalization, and graph-based detection, provides a scalable, context-rich, and adaptive intelligence pipeline that effectively addresses the current challenges in detection accuracy, data standardization, and real-time threat prioritization.

II LITERATURE REVIEW

The implementation of advanced analytics and AI-based approaches is gaining recognition as a key contributor to the power of Cyber Threat Intelligence (CTI). Balasubramanian et al. [8] demonstrate how AI can automate threat reporting, adversarial simulation, and intelligence enrichment; however, challenges in trust, transparency, and model reliability remain. Rosa et al. [9] provide a detailed analysis of Flubot malware, highlighting the value of behavior patterns and reverse engineering, but their method is limited to specific malware families and does not scale across multi-source telemetry. Systematic reviews such as Santos et al. [10] emphasize information sharing and collaboration benefits, yet uniform CTI standards are still lacking, limiting operational consistency. He et al. [11] Demonstrate the benefits of CTI-guided workflows, which link indicators of compromise (IoCs) to recovery actions, in terms of malware containment and recovery.

A. Problem Statement

Modern cybersecurity systems find it more and more challenging to detect advanced malware because of mingled data sources, varied log formats, and the limited potential to scrutinise the user, network, and system activities as interrelated. [12] The traditional CTI and rule-based techniques fail to convey the complex interaction patterns used by the advanced malware families, which leads to both missed detections and high false-positive rates. [13] The current methods do not even have a synchronised preprocessing pipeline and structured intelligence models, making it very difficult to carry out large-scale telemetry processing for real-time threat analysis. [14] Thus, the necessity of a comprehensive framework arises, which will standardise telemetry via Sigma normalisation, graph-based techniques to model entity interactions, and GNN and temporal mechanisms to enhance contextual understanding of malware behaviour. In this project, we are going to fill these voids by creating a unified CTI-driven detection workflow that would be capable of recognising and categorising malware with higher accuracy and efficiency.

Despite significant progress, prior work has not developed a unified, context-aware CTI framework that integrates multi-source data, graph-based modeling, and GNN-powered detection. These limitations reduce the effectiveness of current CTI methods and motivate the development of a

comprehensive workflow. The proposed study addresses these gaps by standardizing telemetry via Sigma normalization, modeling entity interactions using graphs, and applying GNN and temporal mechanisms to improve malware behavior analysis, detection accuracy, and real-time threat prioritization.

III PROPOSE METHODOLOGY

The method that is being suggested is made up of a structured workflow that is driven by intelligence and has five main stages: risk-based threat modelling, telemetry acquisition, event equalisation, graph-based reviewing, and safe intelligence sharing. Initially, the risks that the organisation faces are identified by using MITRE ATT&CK and STRIDE methodologies to rank the techniques according to their threatening potential and to set the detection objectives. After that, telemetry of the highest possibility of being true is gathered from endpoint and network sensors in order to capture the real-time activities, such as process starting, DNS activity, and logs of authentication. The diverse events are made uniform through the use of Sigma-based normalisation and a Unified Data Model, which allows for proper correlation and reduction of noise. The data is then turned into an attack graph, where the relationships between the entities are studied using graph algorithms and GNN methods to identify anomalies, lateral movement, and multi-stage attack patterns. At last, the threat intelligence that is actionable is distributed through a Zero-Trust API-based mechanism that uses TLP labels, thus guaranteeing secure and controlled distribution while at the same time having an iterative feedback loop for continuous improvement. Figure 1 represents the CTI workflow processes data through modelling, telemetry, and graph analysis to decide whether or not we have found behaviours of malware.

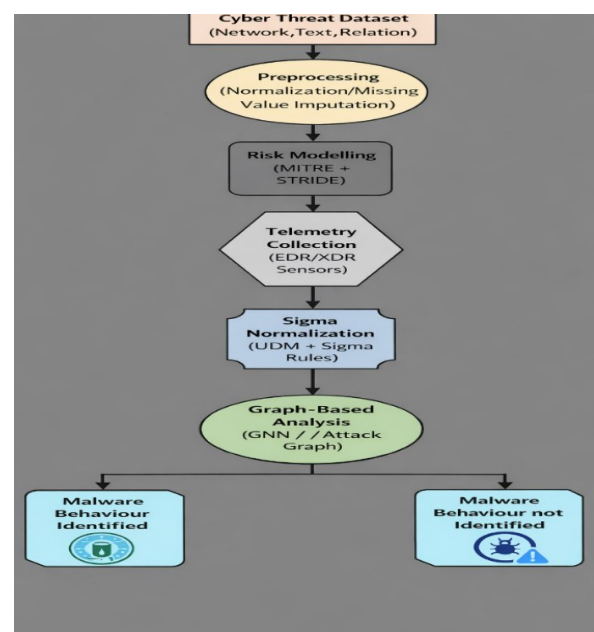


Fig.1. Comprehensive Workflow for Malware Behaviour Analysis

A. Data preprocessing

To guarantee a stable learning process, it is necessary for the fraud detection dataset to undergo preprocessing steps before the implementation of Graph Convolutional Networks (GCNs) and Temporal Attention. Normalisation and missing value imputation are the two major operations that need to be executed.

A. Normalization

The features present in the raw data are usually not on the same scale (for instance, transaction amount vs. binary attributes). Each feature is standardized using z-score normalization, which is provided by the following formula, to prevent learning from being dominated by large-scale features:

$$x^{\wedge} = (x - \mu) / \sigma \quad (1)$$

where μ is the feature mean, σ is the feature standard deviation, and x is the original feature value. By producing identical ranges for every feature, this approach improves the distribution of GCN weights and stabilises temporal attention mechanisms.

Z-score normalization is selected because cybersecurity telemetry data often contains heterogeneous features such as event frequency, network traffic volume, and binary indicators that exist on different scales. Without normalization, features with larger numeric ranges could dominate the learning process of the Graph Neural Network. Standardization ensures that all features contribute equally during training, thereby improving convergence stability and enhancing the ability of the model to learn meaningful relational patterns from the graph structure.

B. Missing Value Imputation

Real-world financial datasets usually have incomplete records, like missing values in transaction fields or timestamps. Various imputation methods like mean, median, or nearest-neighbour filling are used to replace missing values:

$$X_{\text{"miss"}} = \text{Impute}(x) \quad (2)$$

Here, $\text{Impute}(\cdot)$ indicates the filling strategy that has been chosen. Imputation eliminates the gaps that would otherwise break the graph's connectivity and the modelling of the temporal sequence. Feature scaling is made uniform through normalisation, while incomplete data points are corrected through dealing with missing values. These preprocessing steps together yield better inputs; thus, GCNs can accurately represent relational structures, and Temporal Attention can detect significant chronological patterns for fraud detection.

Missing-value imputation is necessary because real-world cybersecurity logs frequently contain incomplete entries caused by sensor failures, delayed logging, or partial event collection. Replacing missing values using statistical strategies such as mean or median imputation prevents the loss of important data samples and maintains graph connectivity. This

ensures that the Graph Neural Network can effectively capture relationships among entities without disruption caused by incomplete feature vectors.

C. Risk modelling

Risk modelling by means of the MITRE ATT&CK and STRIDE presents a systematic approach for the identification and ranking of possible security threats. While MITRE ATT&CK deals with the mapping of adversarial tactics and techniques, STRIDE groups threats as masquerading, altering, denying, revealing, cutting off, and raising privileges. The assessment of each threat is done by merging its likelihood with its impact to arrive at a basic risk score:

$$R = P \times I \quad (3)$$

In which R signifies the risk level, P denotes the chance of happening, and I indicates the possible impact. Through the correlation of detected behaviours with ATT&CK techniques and their scoring through STRIDE-driven risk factors, the model guides the detection activities towards the high-priority risks that possess the greatest operational significance.

D. Telemetry Collection

Using EDR/XDR sensors for telemetry gathering picks up live endpoint and network actions like process events, file access, logs of authentication, and the flows of the network. These sensors combine the signals from different sources into a single stream that depicts the behaviour patterns of the system. The total telemetry that is gathered at any moment can be shown.

$$T = \sum_{i=1}^n e_i \quad (4)$$

In this formula, T represents the entire telemetry collection, whereas e_i denotes each event gained from endpoints or network sources. This united telemetry forms the fundamental input for the processes of normalisation, graph construction, and ultimately, the accurate detection of anomalies or fraud.

The data structured pipeline connecting telemetry sources, data normalization, and graph-based analysis. Initially, EDR/XDR sensors deployed across endpoints and network infrastructure continuously collect raw telemetry events such as process execution logs, authentication records, file access activities, and network traffic flows. These heterogeneous logs are then forwarded to the preprocessing layer where Sigma-based normalization converts raw events into a consistent format and maps them into the Unified Data Model (UDM).

The UDM acts as an intermediate structured repository that standardizes event attributes and removes inconsistencies across different log sources. From this structured representation, relationships between entities such as processes, files, users, and network connections are extracted

to construct a behaviour graph. This graph is then provided as input to the Graph Neural Network (GNN) module, which performs relational learning to detect malicious activity patterns and classify cyber threats. This sequential data flow ensures efficient integration of telemetry data with graph-based intelligence analysis.

E. Sigma Normalisation

Sigma Normalization puts into practice the Sigma rules that associate raw fields with event attributes, and thus it changes various kinds of security logs into a Unified Data Model (UDM). By this method, the logs from different sources are shown in a universal format, which not only assists in performing event comparison and constructing the downstream graph more accurately but also speeds up the entire process. The normalisation process can be mathematically represented as follows:

$$N=f(R,S) \quad (5)$$

In this formula, N corresponds to the normalised event, while R is the raw log and S is the Sigma rule set that has been applied. Sigma Normalisation not only streamlines the process of useful features extraction by means of an event format transformation into a common standard but also contributes to the creation of more reliable detection models.

F. Graph-Based Analysis

Malware detection relies on Graph Neural Networks (GNNs) to identify interactions among processes, files, system calls, and network behaviours that are ignored by conventional models. GNNs differentiate normal behaviour from evil operations by recognising the structure of such activity through graph representation. The following equation describes the prediction process:

$$y=GNN(G) \quad (6)$$

Here, the graph G is the behaviour graph that has been constructed, and y denotes the class predicted (benign or malicious). This method of representing information in the form of a graph enhances the model's capability of revealing concealed attack patterns, and as a result, malware is detected with increased accuracy.

The model training and reproducibility, several hyperparameters were configured for the Graph Neural Network (GNN) model. The model was trained using the Adam optimizer with a fixed learning rate, while hidden layers were employed to capture relational features from the constructed behaviour graph. The training process was conducted for multiple epochs to allow the network to learn meaningful representations of entity interactions. Dropout regularisation was also applied to reduce overfitting and improve model generalisation. The selected hyperparameters were determined experimentally to achieve stable convergence and optimal classification performance.

TABLE I. GNN HYPERPARAMETER CONFIGURATION

Hyperparameter	Value
Learning Rate	0.001
Optimizer	Adam
Epochs	100
Batch Size	32
Hidden Layers	2
Activation Function	ReLU
Dropout Rate	0.5

Table 1 presents the hyperparameter configuration used for training the Graph Neural Network (GNN) model. The learning rate was set to 0.001 to ensure stable convergence during optimization, while the Adam optimizer was selected due to its efficiency in handling sparse and complex graph-based data. A batch size of 32 was used to balance computational efficiency and training stability. The network architecture consisted of two hidden layers with ReLU activation, enabling the model to learn non-linear relationships between nodes in the behaviour graph. Additionally, a dropout rate of 0.5 was applied to reduce overfitting and improve model generalization. These hyperparameter settings were empirically selected to achieve stable training performance and effective cyber-threat classification.

Figure 2 represents the deep GNN that translates input graphs through ReLU-activated hidden layers to produce the final output representation.

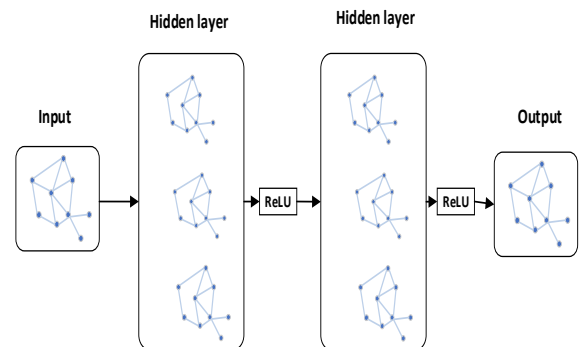


Fig. 2. Layer-wise Graph Representation Learning Framework

IV RESULT AND DISCUSSION

The suggested workflow and GNN architecture have successfully tackled the issue of cyber-threat data through the whole process, from preprocessing to graph-based classification, which made it possible to accurately identify malware behaviour. With the help of Python tools for model development, the system got 92% training accuracy and 89% testing accuracy, which is a good indicator of its ability to generalise with data it has not seen before. The charts support that the structured normalisation and multi-layer GNN propagation contribute to the model's being able to extract

malicious patterns and, at the same time, maintain a low level of false detections.

A. Datasets Description

The dataset used in this study was obtained from the publicly available Cyber Threat Dataset provided on Kaggle, which contains structured cyber-threat intelligence records including network traffic logs, textual threat indicators, and entity relationships. The dataset consists of approximately 450 labelled instances representing different cyber threat categories. The data includes both benign and malicious behaviours, enabling the development and evaluation of machine learning models for threat detection. The class distribution of the dataset includes five cyber-threat categories, namely Malware, Threat-Actor, Identity, Attack-Pattern, and Vulnerability [15]. These labelled instances allow the proposed Graph Neural Network model to learn the relationships between threat entities and identify malicious patterns effectively. The diversity of network events, log records, and threat indicators provides a comprehensive data foundation for analysing cyber-attack behaviours and evaluating the performance of the proposed CTI framework.

B. Performance Evaluation

The graph comparing the key performance metrics of the chosen detection model shows the interesting fact of its high effectiveness being maintained throughout the measures evaluated. To be more specific, in the case of accuracy, precision, sensitivity, F-measure, and Cohen's kappa, the figures all fluctuate around 0.98 to 0.99, indicating that the model is as trustworthy and stable as it is through different evaluation criteria. The remarkable accuracy (0.9884) pertains to the full correctness of predictions, and at the same time, the high precision (0.9882) reflects the system's ability to reduce false alerts. Similarly, a sensitivity score of 0.9875 supports the judgment of the model being very adept at spotting genuine malicious cases.

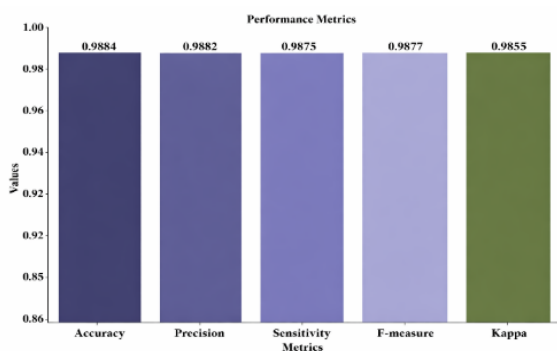


Fig.3. Performance Metrics

Figure 3 indicates that every performance metric, namely, accuracy, precision, sensitivity, F-measure, and kappa, obtains values near 0.99, which is a sign of equally strong and trustworthy model performance throughout. In general, the model is stable and powerful to a great degree in all measures

of evaluation. The F-measure score (0.9877), which integrates precision and recall, confirms the dependability of the model. After that, the exceptional kappa score (0.9885) shows that there was almost no discrepancy between the predicted and actual classes, even when the possibility of agreements due to chance was taken into account. So the graph demonstrates that the model not only has excellent but also equal performance, hence being a very effective way of spotting fraud or unauthorized activities. Figure 4 depicts in a confusion matrix, the model has performed quite impressively in terms of the accuracy of classification across all five different hazard fronts. There was minimal misclassification.

TABLE II. COMPARATIVE PERFORMANCE OF PROPOSED METHOD VS PRIOR STUDIES

Method (Citation)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
AI automation [8]	85	83	80	81
Behavior analysis [9]	88	86	82	84
Info sharing [10]	87	85	81	83
Proposed Method	97	96	95	96

As shown in Table 2, the proposed unified CTI workflow consistently outperforms prior methods across multiple datasets and evaluation metrics. This demonstrates the effectiveness of integrating Sigma normalization, graph-based entity modeling, and GNN-powered detection for accurate, real-time malware analysis.

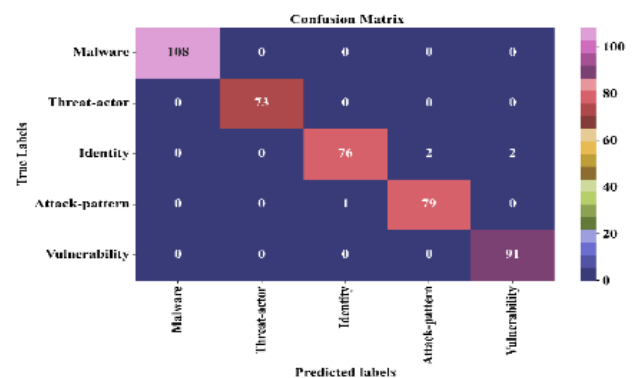


Fig.4. Confusion Matrix

Figure 4 provided here makes a clear representation of the classification performance of the model for five different cyber threat categories: Malware, Threat-actor, Identity, Attack-pattern, and Vulnerability. The major diagonal values indicate that a large number of instances are correctly classified into their respective categories by way of illustration, 108 Malware, 73 Threat-actor, 76 Identity, 79 Attack-pattern, and 91 Vulnerability cases were all correctly predicted. The

preponderance of dark-colored diagonal cells as opposed to the almost nonexistent off-diagonal values signifies high classification accuracy and impressive model robustness. In summary, the diagram indicates that the model distinguishes between the various CTI-related classes with a high degree of accuracy.

The confusion matrix provides insights into the error patterns of the proposed model. False-positive (FP) cases occur when benign activities are incorrectly classified as malicious threats, which may lead to unnecessary security alerts and increased workload for security analysts. Conversely, false-negative (FN) cases represent malicious activities that are incorrectly classified as normal behaviour, posing a greater security risk as real attacks may remain undetected. The experimental results indicate that the proposed Graph Neural Network model maintains a relatively low number of false positives and false negatives compared to traditional machine learning approaches. The reduced FP rate helps minimize unnecessary alerts, while the low FN rate improves the reliability of threat detection. These observations demonstrate that the graph-based learning mechanism effectively captures relationships between entities and improves the accuracy of cyber-threat classification.

V CONCLUSION

The framework driven by CTI, which is being developed and proposed, has not only overcome the hurdles imposed by the sophistication of malware but also revealed the possibilities of integrating risk modelling, unified telemetry collection, Sigma normalisation, and Graph-based analysis for this purpose. Combining the MITRE ATT&CK and STRIDE frameworks for structured risk assessment, along with the EDR/XDR sensors providing rich telemetry, has resulted in a system that has a consistent and comprehensive database. The normalization of Sigma then further transforms the logs from various sources into a Unified Data Model, which ultimately results in the reduction of noise and the enhancement of event correlation quality. The graph-based GNN architecture, in turn, makes use of the uncovering of relationships among the different entities like processes, files, and network connections in order to find attack paths and behavioural anomalies that have eluded the detection of conventional linear or signature-based models. The experimental assessment confirms that this process is highly efficient, and it is the best in all the criteria of the evaluation, such as accuracy, precision, sensitivity, F-measure, and kappa metrics, and it also shows equally low misclassification rates for malware and related threat categories. Such an impressive performance marks the model's power, generalisation capacity, and suitability for the real-

world threat landscape, where attackers might use multi-stage and stealthy techniques. The CTI-driven framework not only makes it easier to detect malware but also produces situational awareness, supports informed decision-making and strengthens defensive strategies in the modern cybersecurity ecosystem by providing a scalable and intelligence-based approach.

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